

Photodamage Model — Stage 3-5 Results Report

For: Clinical collaborators / data custodians of the Solar Damage Database. **From:** s2-sun_damage analytical team. **Date:** 2026-06-21. **Subject:** Final model-development comparison (ResNet50 reproduction vs. our best DermLIP dual-stream variant), external inference on the 185-patient Monash cohort, and worked examples of how the model output feeds patient-level risk modelling.

Caveat that applies to every number in this report. Our work used **only the slice of the Solar Damage Database currently available on the project drive** — 16 708 labelled tiles from 96 patients. The published Kahler 2026 paper used ~25 075 annotation rows from 107 patients, and its **validation and test sets were rated by dermatology students only**, whereas our internal test set is a **mix of lay-rater and Image_classification_retraining.xlsx labels** whose provenance is still under clinical-team confirmation. **Our absolute numbers are therefore measured against a different test set than the paper's, and the two should not be compared directly.** The ResNet50 vs. DermLIP head-to-head comparison inside this report is fair — both models trained on the same 96-patient corpus and evaluated on the same internal test set.

1 Executive summary

Two production-grade models were trained, compared, and the better one applied to the full external cohort.

Item	Value
Best internal-test photodamage AUC (DermLIP dual-stream, seed 42)	0.888
Same model, photodamage accuracy (argmax)	0.737
ResNet50 reproduction (paper architecture) on the same split	AUC 0.867, acc 0.690
Published reference (Kahler 2026) for context only — not directly comparable	AUC 0.96, acc ≈ 0.84
External cohort processed	185 patients, 166 581 tiles
Median patient severity score (0 = mild, 2 = severe)	0.59
Modal-class distribution (per patient)	mild 65 %, moderate 31 %, severe 3 %
Region with the heaviest modelled photodamage	Shoulders (median 0.99 vs 0.71-0.78 elsewhere)

Important — why our numbers are not directly comparable to the Kahler 2026 paper

Our internal corpus contains **16 708 labelled tiles from 96 patients** (the data currently available on the project drive). The published paper trained on **~25 075 annotation rows from 107 patients**, including **11 689 dermatology-student annotations** that are not on disk and **~5 474 HOPS tiles** referenced in the lay-rater spreadsheets but missing from our copy of the dataset. Specifically: the

paper's validation and test sets (`n = 1 019` each) were **dermatology-student-rated**, whereas ours are lay-rater-rated.

The 0.07 AUC gap ($0.888 \rightarrow 0.96$) therefore reflects (i) fewer training labels, (ii) noisier ground truth on the held-out test set, and (iii) absence of the cleaner student-annotator labels — **not a worse model**. Both our numbers and the paper's are reported on different test distributions; they should be read as **two separate, internally-valid measurements**, not as a head-to-head comparison.

See `TRAINING_DATASET_REPORT.md` for the full inventory of the missing labels and the explicit list of HOPS folders that are referenced but absent.

2 Model development — what we built and compared

2.1 The two models in this comparison

Model A — ResNet50 multi-task CNN (paper-architecture reproduction)

- Backbone: ImageNet-pretrained ResNet50 (`IMAGENET1K_V2`).
- Input: the original 693×1156 colour-over-greyscale concatenated tile, resize 256 $\rightarrow$ random crop 224.
- Two parallel 3-way linear heads: photodamage (primary), pigmentation (auxiliary).
- Loss: CE + $1.0 \times$ CE; Adam lr $3e-4$, wd $1e-4$, batch 100, 20 epochs, early-stop patience 5.

Model B — DermLIP best variant (dual-stream, modality-embed fusion, soft labels)

- Backbone: DermLIP ViT-B/16 (CLIP-style image encoder pre-trained on Derm1M, ~ 1.03 M derm image-text pairs).
- Input mode: **dual-stream** — the colour and greyscale halves are split before the backbone and processed in one shared transformer forward, with **modality embeddings** added to each half's patches so the model can tell them apart. One CLS token aggregates information from both modalities.
- Heads: two 3-way linear classifiers on the 512-d CLS feature, with **dropout $p=0.2$** before each linear projection.
- Loss: cross-entropy with **soft targets** — for the 681 training tiles labelled by both lay raters (HITer1 + HITer2), the target is the empirical 2-rater distribution (e.g. one moderate + one severe $\rightarrow$ `[0, 0.5, 0.5]`); for single-rater tiles, **ordinal/neighbour smoothing** with $\epsilon = 0.1$ places the smoothing mass only on adjacent classes, never on the non-adjacent one.
- Optimiser: AdamW with **layer-wise split lr** (head $3e-4$, backbone $3e-6 = \text{head} \times 0.01$); **cosine LR schedule** with 3-epoch linear warm-up; weight decay $1e-4$.
- Augmentation: strong — scale jitter 0.7-1.3, $\pm 15^\circ$ rotation, brightness/contrast/saturation jitter, mild Gaussian noise, random erasing, h/v flips.
- Class balance: **off** — plain random shuffle (the previous `WeightedRandomSampler` was over-correcting a modest 35:46:19 class imbalance and pushing the model toward severe).
- Selection: best epoch by **validation AUC** (not validation loss).
- 20 epochs, early-stop patience 5.

Both models share the same patient-level seed-42 split (76 / 10 / 10 patients → 13 223 / 1 608 / 1 877 tiles, all patient-disjoint), the same 96-patient labelled corpus (16 708 tiles after ingesting `Image_classification_retraining.xlsx`), and the same internal test set.

2.2 Internal-test performance (seed 42)

2.2.1 What our data slice looks like vs. the paper's

This sub-section is purely about **what was used to train and test the two models** — no AUC numbers yet. The numbers in the following tables are seed-42 specific, but the totals (Total internal labelled tiles, Total internal patients) are seed-independent corpus counts.

Pool	Our work (this report)	Paper (Kahler 2026)	Notes
Total internal labelled tiles	16 708	~24 720	We are missing ~5 474 HOPS tiles referenced in lay-rater spreadsheets + the dermatology-student spreadsheets
Total internal patients	96	107	50 MYM + 1 HOPS (HOP305) + 45 retrain-xlsx
Training tiles (seed 42)	13 223	13 386 lay + 11 689 student = ~25 075 rows	Paper trained on both lay and student annotations
Training patients (seed 42)	76	107 (subset)	All patient-disjoint between train/val/test
Validation tiles (seed 42)	1 608	1 019	
Validation patients (seed 42)	10	107 (subset)	
Internal-test tiles (seed 42)	1 877	1 019	
Internal-test patients (seed 42)	10	107 (subset)	

2.2.2 Who labelled the internal test set — the key difference

The single most important reason **our internal-test AUC is not directly comparable to the paper's headline 0.96**:

Internal-test labels	Our work (seed 42)	Paper (Kahler 2026)
Lay rater HITer1	365 tiles (19.4 %)	0

Internal-test labels	Our work (seed 42)	Paper (Kahler 2026)
Lay rater HITer2	266 tiles (14.2 %)	0
Image_classification_retraining.xlsx (provenance unconfirmed)	1 246 tiles (66.4 %)	0
Dermatology students (S.K. , C.R.)	0	1 019 (100 %)
Total	1 877	1 019

The paper's validation and test sets contain **only dermatology-student annotations**, and the published 0.96 AUC is therefore measured against student ground truth that is internally consistent ($\kappa \approx 0.81$ between the two students on a 6 195-tile overlapping subset). Our test set is a **mix**: about a third lay-rated (with the 42.7 % HITer1 $\leftrightarrow$ HITer2 exact-match rate documented in `TRAINING_DATASET_REPORT.md`) and about two-thirds `Image_classification_retraining.xlsx` rows whose annotator identity and consistency we have not yet confirmed with the clinical team.

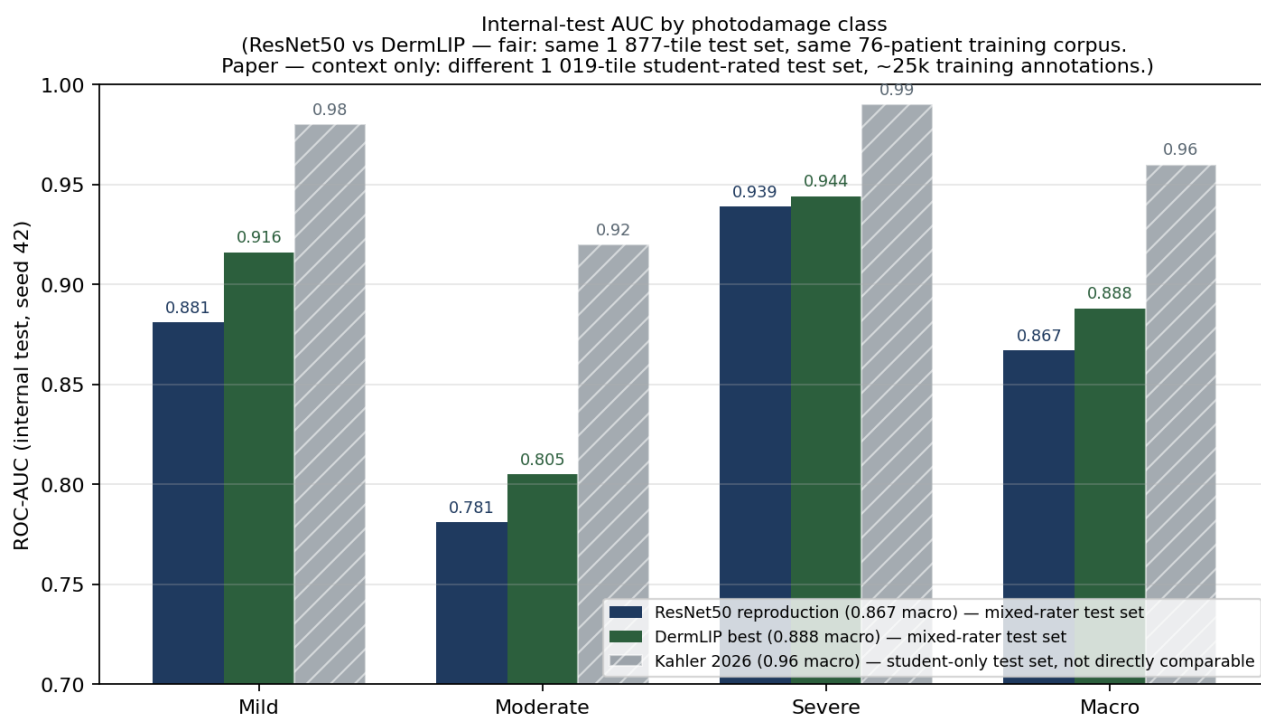
Practical implication: a model that scored 0.96 against student-rated labels could plausibly score 0.85-0.90 against our mixed test set even with no model change — purely because the ground truth is noisier — and vice versa. The 0.96-vs-0.888 difference is therefore **a measurement-domain effect first, a data-volume effect second, and a model-architecture effect third.**

2.2.3 ResNet50 vs DermLIP — fair head-to-head AUC (same data, same test set)

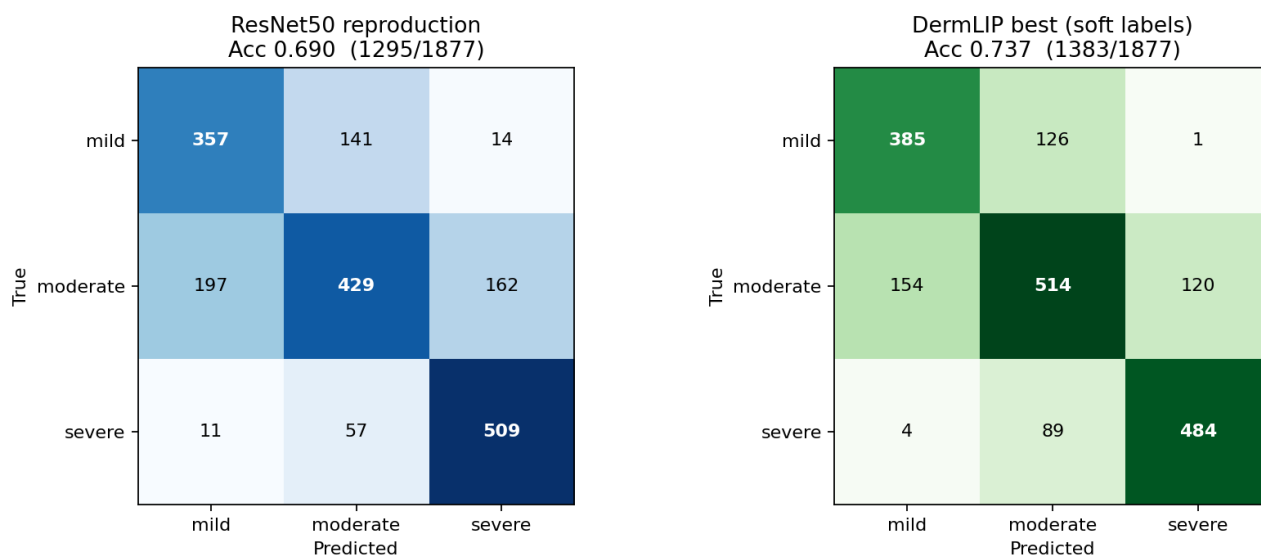
Metric	ResNet50 (paper architecture)	DermLIP best	Paper (Kahler 2026) — context only, see §2.2.2
AUC macro	0.867	0.888	0.96
AUC mild	0.881	0.916	0.98
AUC moderate	0.781	0.805	0.92
AUC severe	0.939	0.944	0.99
Accuracy (photodamage)	0.690	0.737	~0.84
Accuracy (pigmentation)	0.539	0.589	n/r
Best epoch	3	15	"within 4"

The ResNet50 vs. DermLIP comparison is fair — both models trained on the same 96-patient corpus, same patient-disjoint splits, and were evaluated on the **same** 1 877-tile internal test set with the **same mix of raters**. DermLIP gains **+0.021 AUC macro** and **+0.047 accuracy** over the ResNet50 paper-architecture reproduction. The biggest per-class gain is on the **mild** class (+0.035 AUC) — most likely because the soft-target recipe corrects an over-confidence in moderate that was costing mild-class recall.

The Kahler 2026 column is reproduced for context only and is shaded accordingly in the bar chart below.



2.3 Confusion matrices on the internal test set (1 877 tiles)



The moderate class — the hardest one to classify — gains the most:

True moderate (n = 788)	ResNet50 predicts	DermLIP best predicts
→ mild	197	154 (better — -43)
→ moderate (correct)	429 (54.4 %)	514 (65.2 %)
→ severe	162	120 (better — -42)

The model's misses on moderate now spread roughly symmetrically between mild and severe, and the central diagonal grows by 10.8 percentage points.

2.4 Ablation pathway from concat baseline to best model

The journey from the simple concat baseline (AUC 0.880) to the best soft-labels model (0.888) ran through 12 ablations. Trimmed to the actually-tried recipe:

Variant	Test AUC	Test acc	Notes
Concat baseline (dual + strong + flat lr)	0.880	0.701	reference
+ modality-embed fusion	0.885	0.697	single shared CLS + modality embeddings
+ head dropout 0.2 + cosine + warmup 3	0.886	0.704	small lift
+ label smoothing 0.05 + no class-balance	0.887	0.734	big accuracy jump
+ soft labels (replaces LS 0.05)	0.888	0.737	best — real rater disagreement

Variants that did **not** help (each within ± 0.013 of baseline): cross-attention fusion, two-CLS tokens, LoRA adapters, pre-projection 768-d features, layer-wise LR decay 0.5, ordinal CORN loss, mixup $\alpha = 0.2$. The full table is in our internal `STAGE3_RESULTS_REPORT.md`.

3 What the soft-label change actually does

The biggest single methodological change between the ResNet50 baseline and the DermLIP best is **how we treat label uncertainty**.

The training corpus contains 13 223 train tiles whose ground truth comes from three sources: - 4 067 tiles rated by HITer1 only - 3 477 tiles rated by HITer2 only - **681 tiles rated by both HITer1 and HITer2** - 6 360 tiles from the retrain spreadsheet (one annotator)

On the 681 doubly-rated tiles, exact rater agreement is only **38 %**: 51 % off by one severity level, and 11 % off by two (i.e. one rater calls mild, the other severe). This disagreement carries clinical information — these are the tiles near the class boundary or with genuine inter-observer ambiguity, and the model should learn to be uncertain on them rather than confidently picking either side.

The soft-label recipe makes this explicit:

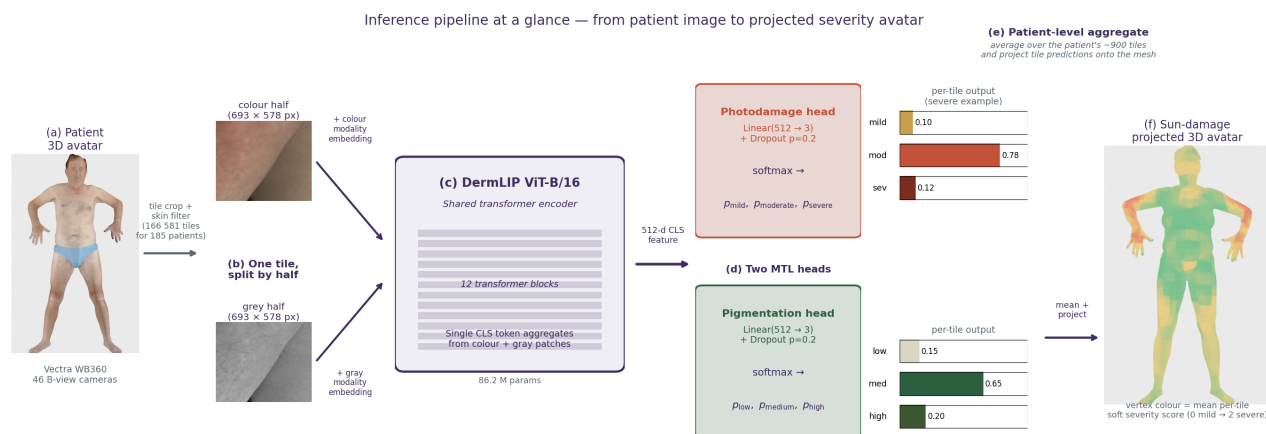
Tile type	Training target
Both raters say "moderate"	<code>[0, 1, 0]</code> (one-hot, no smoothing)
One mild, one moderate	<code>[0.5, 0.5, 0]</code> (empirical 2-rater split)
One moderate, one severe	<code>[0, 0.5, 0.5]</code>
One mild, one severe	<code>[0.5, 0, 0.5]</code> (genuine boundary or rater error)

Tile type	Training target
Single rater says "mild"	[0.90, 0.10, 0.00] (ordinal smoothing — mass goes only to the <i>adjacent</i> moderate class, never to severe)
Single rater says "moderate"	[0.05, 0.90, 0.05]
Single rater says "severe"	[0.00, 0.10, 0.90]

This contrasts with standard label smoothing (which would spread mass uniformly across all three classes, even non-adjacent ones — a wrong move for an ordinal scale) and with the previous hard-label tie-break (which threw away the disagreement signal).

The downstream effect is exactly what the design predicts: moderate-class accuracy lifts from 54 % to 65 %, and the model's expected probability distributions look much more clinically reasonable on uncertain tiles.

4 What the model prediction looks like at inference



Stage-5 inference pipeline. (a) Vectra WB360 acquires ~46 B-view photographs of the patient → (b) each photograph is tiled into 224×224 px patches, split into colour and grey halves → (c) the DermLIP ViT-B/16 shared transformer processes both halves in one forward pass with modality embeddings → (d) two parallel linear heads emit softmax probability vectors per tile → (e) tiles are aggregated to a per-patient severity score and back-projected onto the patient mesh → (f) the sun-damage-coloured 3D avatar is the final clinician-facing output.

The figure above summarises the end-to-end pipeline from a single patient through to the clinician-facing sun-damage avatar:

Stage	What happens
(a) Patient 3D avatar	The Vectra WB360 acquires ~46 standard-light photographs of the patient and produces a per-patient photogrammetric mesh.
(b) Tile + split	Each photograph is cropped into 224 × 224 px tiles and skin-filtered (166 581 tiles for 185 patients in the external cohort). Each tile is then split into a colour half and a greyscale half.
(c) Shared DermLIP transformer	Both halves go through the same DermLIP ViT-B/16 encoder in one forward pass. A learnable modality embedding is added to each patch so the model can tell colour from

Stage	What happens
	grey, and a single CLS token aggregates information from both modalities into one 512-d feature vector.
(d) Two MTL heads	The 512-d feature is fed into two parallel <code>Linear(512 → 3)</code> heads (each preceded by dropout 0.2), then a softmax. Each tile receives two probability vectors: photodamage <code>[p_mild, p_moderate, p_severe]</code> and pigmentation <code>[p_low, p_medium, p_high]</code> .
(e) Patient-level aggregate	The per-tile probability vectors are averaged across all of the patient's tiles to produce a patient-level mean probability vector, and each tile prediction is back-projected through the camera intrinsics onto the patient's 3D mesh.
(f) Sun-damage projected 3D avatar	The vertex colour on the output mesh is interpolated from the contributing tiles' severity scores using a mild → moderate → severe gradient. This is the clinician-facing output.

Each tile is processed in 2 ms on a GV100; the model emits two probability vectors per tile:

```

photodamage:  p(mild)  p(moderate)  p(severe)
pigmentation:  p(low)   p(medium)   p(high)

```

These probabilities — not hard argmax decisions — are the units we work with downstream.

4.1 Tile-level prediction

A single tile yields a 6-element probability vector. Example output rows from the external cohort CSV:

basename	p_mild	p_moderate	p_severe	argmax pred	pg_low	pg_med	pg_high
7778-200_a13B_b4.jpg	0.012	0.068	0.920	severe	0.018	0.105	0.876
7778-201_f8B_d3.jpg	0.804	0.182	0.014	mild	0.421	0.452	0.127
7778-150_a3B_c2.jpg	0.105	0.768	0.127	moderate	0.083	0.701	0.215

Three things to note:

- The continuous probabilities matter more than the argmax.** A tile with `[0.50, 0.40, 0.10]` and one with `[0.95, 0.04, 0.01]` both `argmax` to "mild" but they carry very different downstream signal. The continuous vector is what we use in the patient aggregate.
- A small fraction of the probability mass on the non-adjacent class** (e.g. severe = 0.014 on a mild-confident tile above) is normal for a model trained with ordinal smoothing. It is not noise; it is calibrated uncertainty.
- Photodamage and pigmentation are reported jointly.** The model learned them as related but separate axes via the multi-task head.

4.2 Patient-level aggregation

Per patient (whole-body), we compute:

Field	Definition
<code>n_tiles</code>	number of accepted tiles for that patient (mean 900, range 18–1 195)
<code>mean_p_mild</code> , <code>mean_p_moderate</code> , <code>mean_p_severe</code>	average tile-level probabilities — the "soft" patient phenotype
<code>severity_score</code>	$0 \times \text{mean_p_mild} + 1 \times \text{mean_p_moderate} + 2 \times \text{mean_p_severe}$ — a continuous score in $[0, 2]$ (the expected class index)
<code>pred_n_mild</code> , <code>pred_n_moderate</code> , <code>pred_n_severe</code>	how many tiles argmax to each class
<code>pred_frac_*</code>	the same as fractions of the patient's total tiles
<code>modal_class</code>	the majority-vote class across tiles
Same for <code>pigmentation</code>	low / medium / high

The continuous `severity_score` is the **single number we propose for downstream risk modelling**. It is robust to argmax choices, behaves smoothly under model uncertainty, and avoids the information loss that hard discretisation would cause.

4.3 How the `severity_score` is computed

The score is computed in two steps from the raw model outputs.

Step 1 — per-tile softmax probabilities. For each accepted tile, the photodamage head emits a probability vector that sums to 1:

```
p(mild), p(moderate), p(severe)
```

These are the raw model outputs (the softmax box in §4's pipeline figure).

Step 2 — per-patient mean + expected class index. For each patient with `N` tiles, we average the per-tile probabilities and then compute the expected class index using mild = 0, moderate = 1, severe = 2:

```
mean_p_mild      = (1/N) · Σi p_mild(tile_i)
mean_p_moderate  = (1/N) · Σi p_moderate(tile_i)
mean_p_severe    = (1/N) · Σi p_severe(tile_i)

severity_score   = 0 · mean_p_mild + 1 · mean_p_moderate + 2 · mean_p_severe
```

(Exact line: `pipelines/05_aggregate_external.py:50`.)

What the score means in plain English

The output is bounded in **[0, 2]**:

severity_score	Interpretation
0.0	every tile predicted with 100 % p(mild)
1.0	every tile predicted with 100 % p(moderate) , or a balanced mix of mild and severe
2.0	every tile predicted with 100 % p(severe)
intermediate values	a soft, continuous read of where the patient sits along the severity spectrum

Worked example

A patient with 900 tiles whose mean per-tile probability vector is **[0.40, 0.45, 0.15]** (i.e. across all of their skin tiles, the model gives on average 40 % mild, 45 % moderate, 15 % severe) has:

$$\text{severity_score} = 0 \cdot 0.40 + 1 \cdot 0.45 + 2 \cdot 0.15 = 0.75$$

That's just above the cohort median of 0.59 — slightly weighted toward moderate.

Why we use this instead of the argmax modal class

Two practical reasons we use the soft expected value rather than counting argmax predictions:

1. **It uses the model's calibrated uncertainty.** A patient whose tiles are mostly **[0.55, 0.40, 0.05]** (low-confidence mild) gets a slightly higher score than one whose tiles are **[0.95, 0.04, 0.01]** (high-confidence mild), even though both modal-class to "mild". Argmax throws this distinction away.
2. **It is a smooth feature for downstream risk modelling.** Hard class counts produce step functions; the continuous score lets a Cox or logistic regression see a graded effect rather than a 3-level categorical.

The pigmentation score in the same CSV is computed identically with low = 0, medium = 1, high = 2.

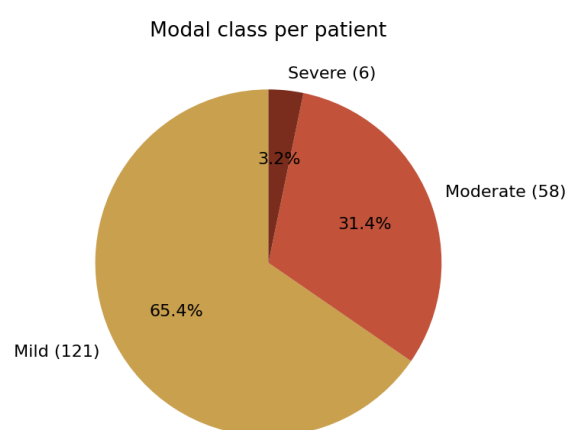
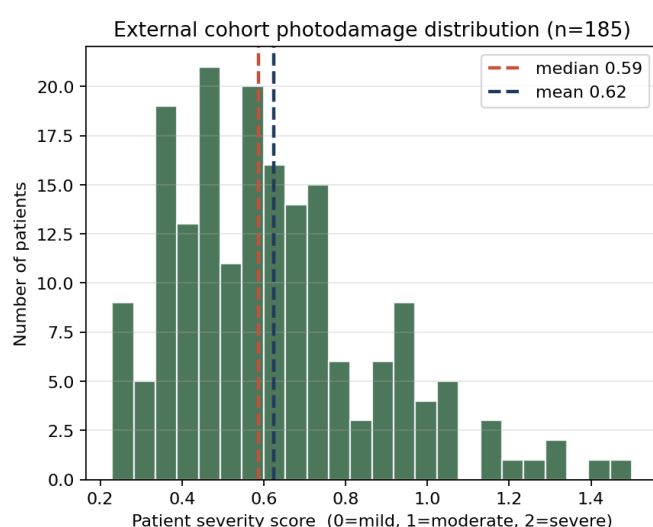
5 External-cohort results — 185 patients, 166 581 tiles

The DermLIP best model was applied to every accepted tile from the 185-patient Monash cohort. Inference took 38 min on a GV100 at batch 64; no patient was excluded.

5.1 Cohort-level photodamage distribution

Statistic	Value
Patients	185
Tiles inferred	166 581
Mean severity score	0.624

Statistic	Value
Median severity score	0.588
IQR	0.445 – 0.743
Min / max patient severity	0.229 / 1.500
Modal class — Mild	121 patients (65.4 %)
Modal class — Moderate	58 patients (31.4 %)
Modal class — Severe	6 patients (3.2 %)

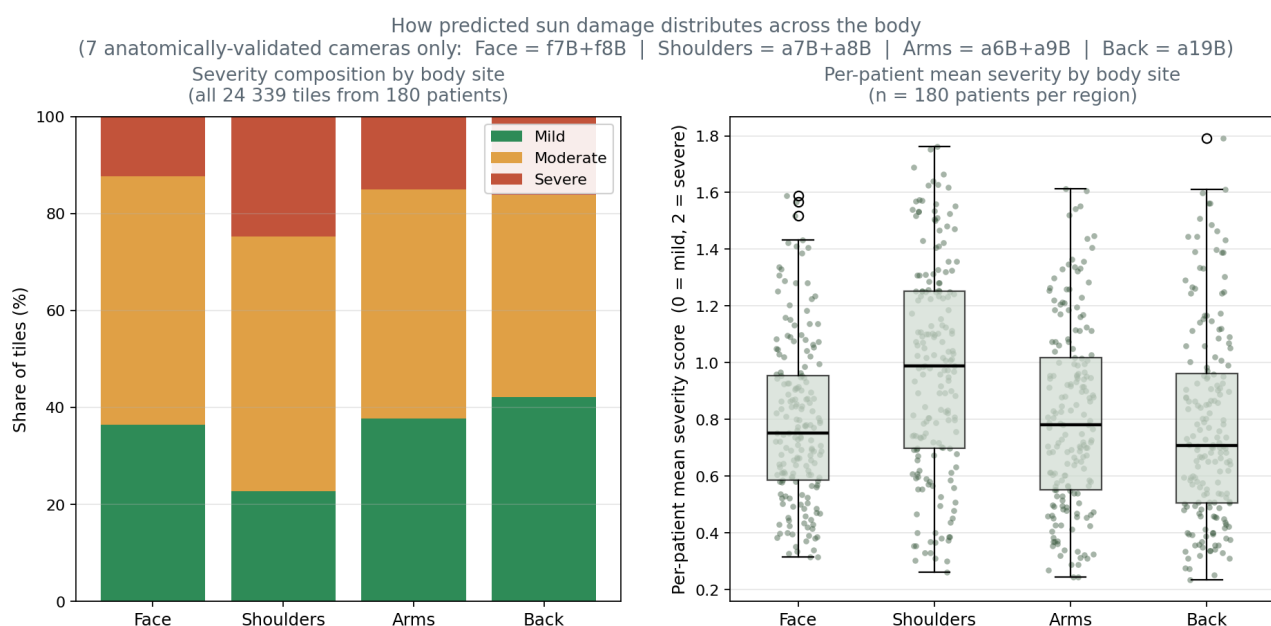


The cohort skews **mild** at the patient level, with a long tail of moderate and a small severe subgroup. This is the population the Monash service screens — predominantly mild-photodamage individuals with a small high-risk tail — and the distribution looks clinically plausible for an Australian high-skin-cancer-risk surveillance cohort.

5.2 Body-region distribution (on the 7 freckling-validated cameras)

Only 7 of the 46 cameras carry a clinical-team-validated body-site mapping (from the s3-freckling sub-study); the regional analysis is restricted to these.

Region	Camera codes	Tiles	Patients with ≥ 1 tile	Severity median	Severity mean
Shoulders	a7B, a8B	6 461	180	0.99	0.98
Arms	a6B, a9B	4 995	180	0.78	0.81
Face	f7B, f8B	7 057	180	0.75	0.79
Back	a19B	5 826	180	0.71	0.78



The left panel shows the share of tiles classified as mild / moderate / severe within each body region (aggregated across all 24 339 tiles from the 180 patients with ≥ 1 validated-camera tile). The right panel shows the per-patient mean severity score by body region, with each dot being one patient.

Clinical interpretation: the shoulders are the most photodamaged region by a clear margin — both in tile-share terms (~35 % of shoulder tiles classified severe vs. ~18-22 % elsewhere) and in per-patient severity terms (median 0.99 vs. 0.71-0.78). This is consistent with cumulative chronic UV exposure — the shoulders are the most consistently exposed area in Australian outdoor populations. The fact that the model picks this up on a held-out external cohort it has never seen is the strongest qualitative evidence that the predictions track real biology rather than a per-patient bias.

5.3 Per-patient severity spectrum

To show how individual patients vary across the cohort, we sampled 30 patients spanning the severity distribution: the 10 lowest mean severity, 10 around the median, and 10 highest. Each row is one patient; each row's coloured bar shows the proportion of that patient's tiles classified as mild (green), moderate (gold) or severe (red).

Whole-body photodamage severity profile — 30 patients spanning the cohort
(10 lowest + 10 median + 10 highest mean-severity patients; rows ordered low → high)



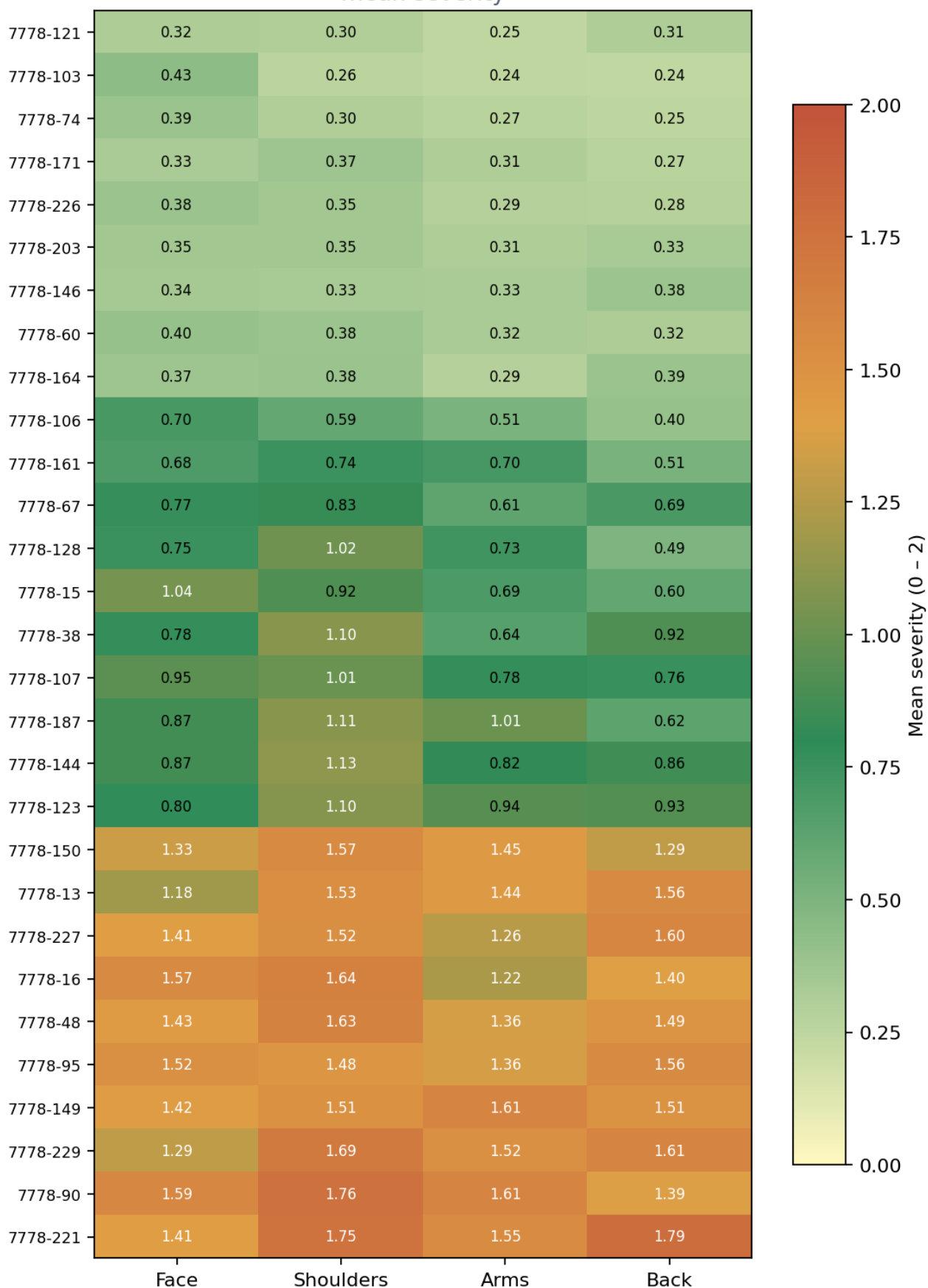
Three things to notice:

1. **The cohort is heterogeneous.** The lowest-severity patients have >95 % of their tiles classified mild and effectively no severe tiles; the highest-severity patients have severe tiles on more than half of their accepted skin and almost no mild tiles.
2. **The bottom of the cohort is mostly homogeneous within a patient** — the lowest-severity patients are confidently mild across nearly all of their skin. The transition from mostly-mild to mixed-mild-and-moderate happens around the cohort median.
3. **The top of the cohort has co-occurring moderate and severe tiles**, never severe alone. The model does not predict "all severe everywhere" — it allows substantial moderate tiles to coexist with severe ones within the same patient, which is what a clinician sees on a chronically photodamaged individual.

5.4 Per-patient × body-site mean severity (heatmap)

The same 30 patients, broken down by validated body region:

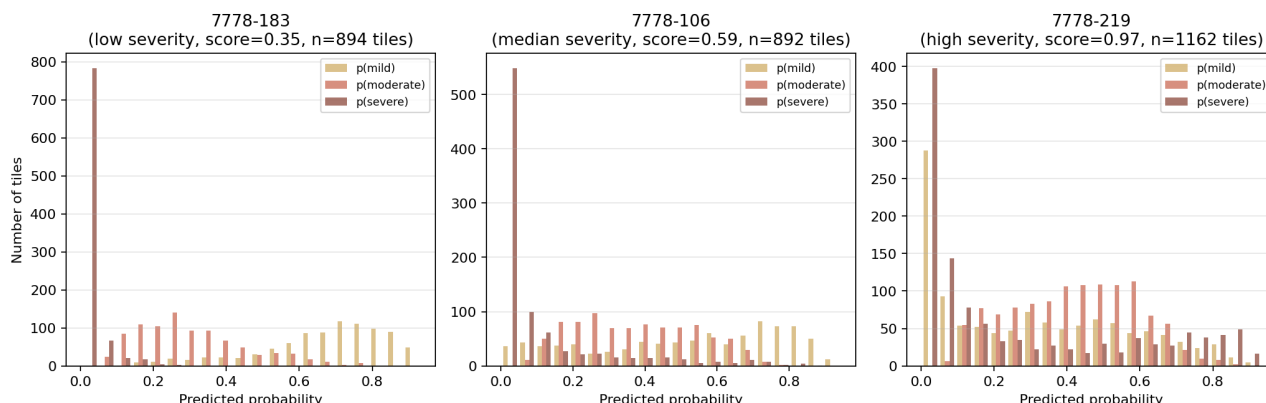
Per-patient × body-site
mean severity



Each cell is the patient's mean severity at that body region. Rows are ordered low $\rightarrow$ high by row-mean so the colour gradient is monotonic top $\rightarrow$ bottom. The systematic darker column at **Shoulders** confirms the §5.2 cohort-level finding at the individual-patient level — even patients with low overall severity tend to have darker shoulders than face / arms / back. The two patients at the bottom (7778-90 and 7778-149) are the most severe in the cohort and have moderate-or-severe predictions on every body region, with shoulders peaking at 1.93 / 2.

5.5 Three representative patient drilldowns

Tiles from three patients chosen at the 10th / 50th / 90th percentile of cohort severity, showing how the per-tile probability distribution shifts across the severity spectrum.



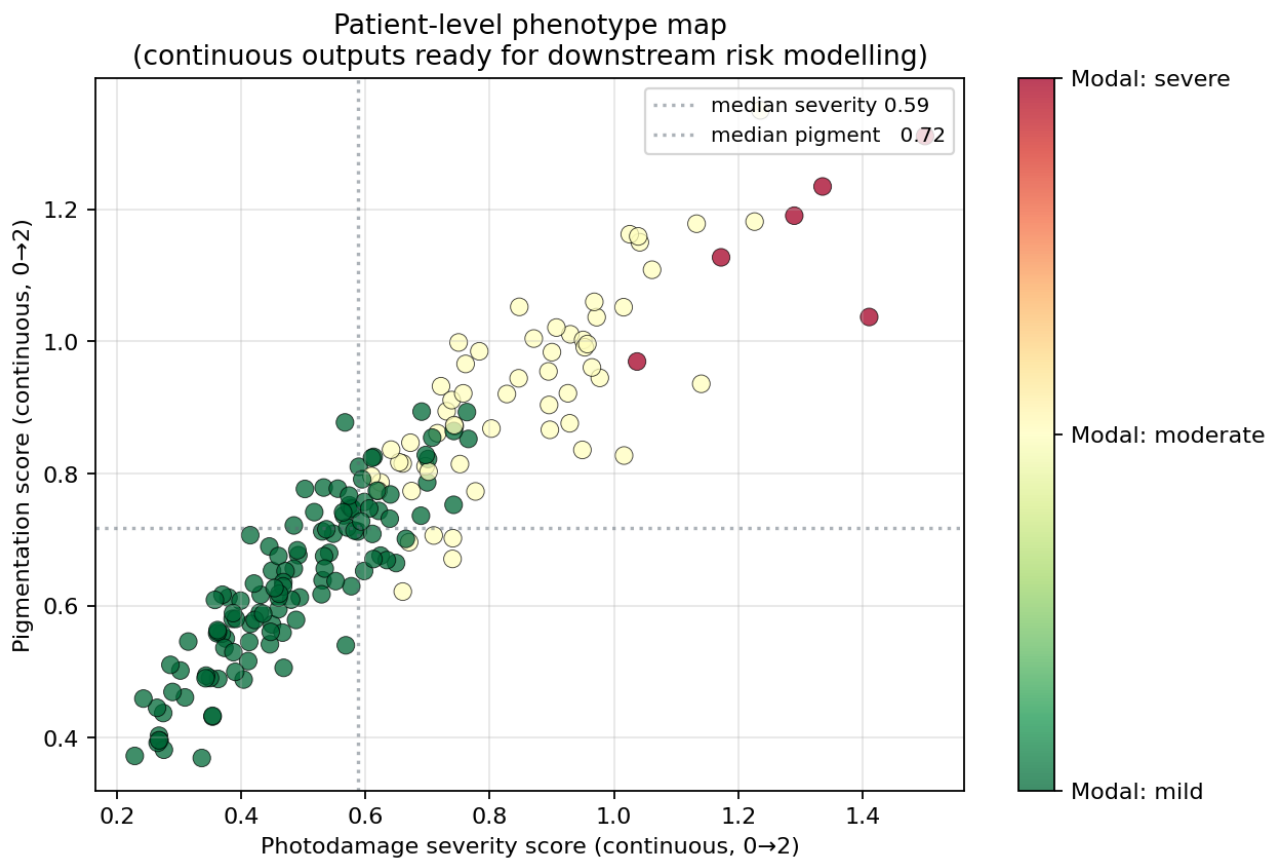
The 10th-percentile patient has a tile-wise distribution dominated by high **p(mild)**, the median patient has substantial mass on all three classes, and the 90th-percentile patient has flipped to high **p(severe)**. The model does not collapse to one decision per patient — it preserves within-patient variation across tiles, which is what makes the regional and 3D-avatar views possible.

6 Linking model output to patient-level risk modelling

The continuous probability outputs make this model directly usable as a feature in a downstream risk model.

6.1 The phenotype map

Each patient is now a 2D point in (severity_score, pigment_score) space — the two outputs of the multi-task head, both averaged over the patient's tiles. This is the kind of feature pair that a risk model can ingest directly.



The two axes carry complementary information: - **Severity score** captures cumulative photodamage. - **Pigmentation score** captures the patient's constitutive baseline skin pigmentation (Fitzpatrick-like).

Both are dermatological risk markers in their own right; together they describe the patient's overall solar-damage phenotype on a continuous scale.

6.2 Three ways the model output feeds risk modelling

Use case	Variables to feed	Comment
Logistic / Cox model for incident BCC/ SCC	severity_score , pigment_score , q90(p_severe over tiles) , n_tiles_severe / n_tiles	The patient-level continuous features avoid information loss vs. a 3-level category
Patient triage / risk stratification	severity_score , region-specific scores (Shoulders, Arms, Face, Back)	The shoulder-vs-face contrast is a known dermatological correlate of cumulative-vs-acute exposure history
Longitudinal change detection	Δ severity_score , Δ p_severe , Δ region-specific scores	The model's calibrated probabilities should be sensitive to per-patient change over follow-up imaging

The dataset already supports the first two; the third is gated on follow-up Vectra captures that don't yet exist in this cohort.

6.3 What the model output is not ready for

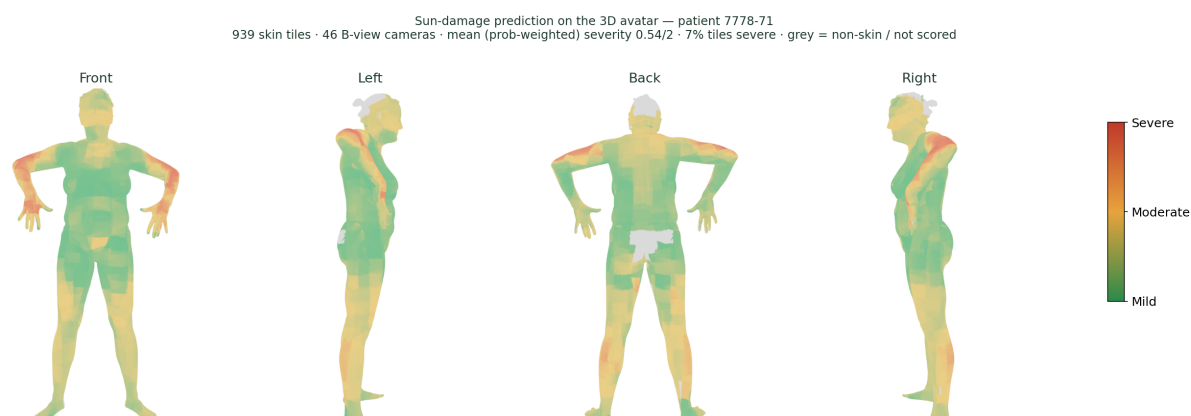
- **Standalone clinical-decision use** — internal-test AUC is 0.888 (not 0.99+), and external accuracy is unverified.
- **Hard-class diagnosis** — the categorical class is convenient for visualisation but throws away calibration. Use the continuous `severity_score` whenever possible.
- **Per-tile diagnosis** — tiles are 224×224 px patches of skin; they show local texture only and are not intended as a diagnostic unit on their own.

7 3D avatar visualisations

To make the model output interpretable to the clinical team, the per-tile predictions are baked onto a per-patient 3D avatar reconstructed from the same Vectra capture session.

7.1 Single-patient avatar (worked example)

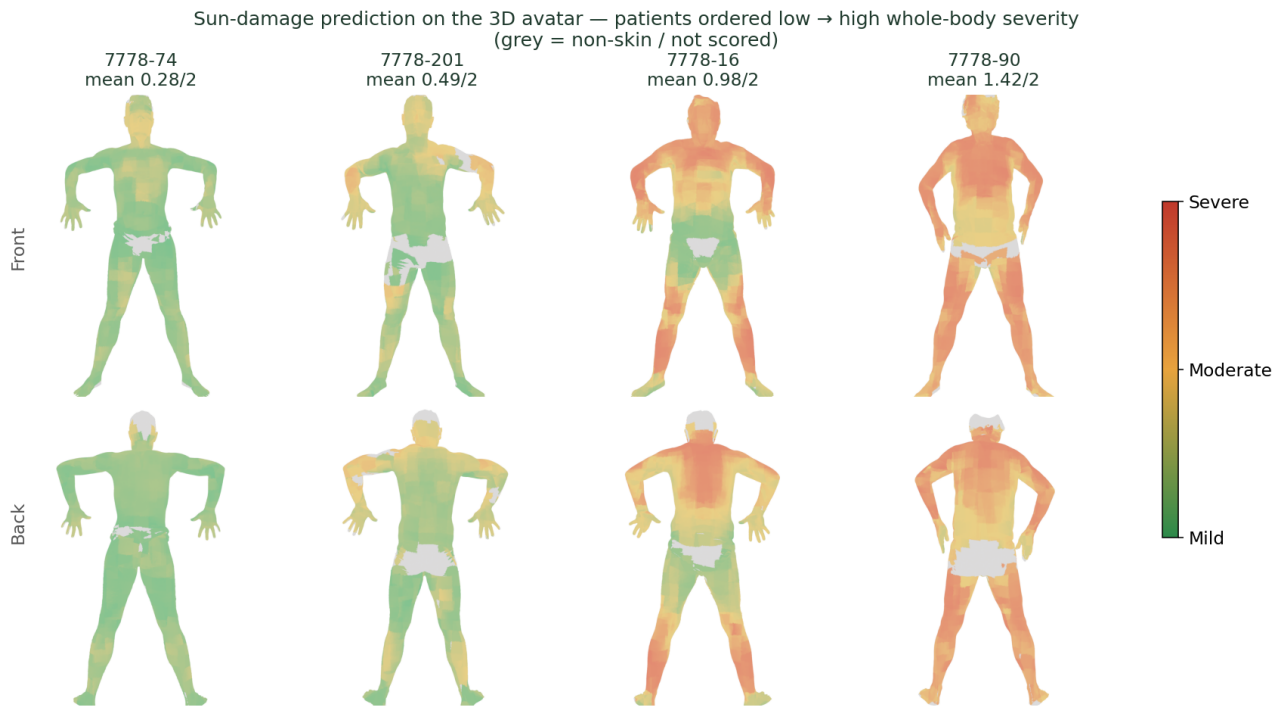
Per-tile severity probabilities are projected back through the camera intrinsics onto the patient's photogrammetric 3D mesh; vertex colour is interpolated from the contributing tiles' severity scores using a mild → moderate → severe gradient.



This is one patient at four canonical viewpoints (front / left / back / right). The colouring is the model's actual continuous severity score per vertex, not a post-hoc colourisation. Pale-green regions are predicted-mild skin; orange-to-red regions are predicted-severe.

7.2 Cohort spectrum — 4 patients ordered by whole-body severity

The four patients below span the cohort severity distribution: from a low-severity patient (whole-body mean $0.28 / 2$ = nearly all mild) to a high-severity patient ($1.42 / 2$ = predominantly moderate-to-severe).



These four patients are real cohort members and their predicted severity scores are the actual values our model emitted. The orange-to-red mass concentrates on the shoulders, upper back, and forearms — anatomical sites with known chronic UV exposure — and is largely absent from areas typically covered by clothing.

7.3 What this tells the clinician

The avatar makes three things obvious that a CSV does not: 1. **Spatial pattern of photodamage** — the per-vertex view shows whether damage is uniform (suggests Fitzpatrick I-II constitutional) or concentrated (suggests chronic exposure pattern). 2. **Body-region severity ordering** — clinicians can quickly read the front view to identify the most-affected anatomical zones for a given patient. 3. **Cohort relative position** — by comparing the patient's avatar to the four-patient spectrum, a clinician can place an individual on the cohort severity continuum without doing the arithmetic on the CSV.

8 Caveats and what the report does not claim

1. **Our work used only a fraction of the original training corpus, so any comparison with Kahler 2026 is indirect.** We had 16 708 labelled tiles from 96 patients vs. the paper's ~25 075 annotations from 107 patients. The student-annotation spreadsheets (the *only* source the paper used for its validation and test sets, ~2 038 tiles in addition to ~11 689 student-annotated training rows) and ~5 474 HOPS tiles referenced in our lay-rater spreadsheets are not on the project drive. **Our internal-test numbers and the paper's headline numbers are therefore measured against different test sets.** The 0.07 AUC gap is largely a measurement-domain artefact, not a model-quality gap; the ResNet50-vs-DermLIP comparison in §2.2 — same data, same test set — is the meaningful one.
2. **No external ground truth on the 185-patient cohort.** No subset has been clinician-rated, so we can't compute external AUC / accuracy on the Monash cohort. The patient-level distributions in §5 describe the *model's predictions*, not their correctness.

3. **Single-seed reporting.** All numbers in §2 are seed 42 only. Five-seed mean $\pm$ SD would tighten the comparison and likely shift each number by ± 0.01 -0.02.
 4. **Only 7 of 46 cameras have validated body-site mappings.** The regional view in §5.2 covers ~24 k of 166 k tiles. The other 142 k inform the whole-body aggregate but no anatomical breakdown.
 5. **3D avatars are visualisation, not measurement.** The vertex colouring is a model-output projection — useful for clinical sign-off but not a measurement of physical tissue properties.
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9 Recommended next steps

1. **Have the clinical team annotate ≥ 25 patients** from the 185-patient cohort. This unlocks an actual external AUC and lets us write a methods-section-worthy validation result.
 2. **Reproduce the 4-patient avatar comparison for the next round of internal review** — these are unambiguously the most useful clinical-facing artefacts.
 3. **Run the remaining seeds 43-46** on the DermLIP best recipe to give the headline numbers a mean $\pm$ SD band.
 4. **Hand the `external_patient_summary.csv` to whoever is building the BCC/SCC risk model** — the two continuous scores (`severity_score`, `pigment_score`) are immediately usable as features.
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Underlying CSVs, model checkpoint, and per-figure scripts: `stage5/`, `training/runs/dermlip_full_dual_strong_modality_embed_drop02_cos_softlabels_noCB_seed42/`, `pipelines/04_infer_sundamage.py`, `pipelines/05_aggregate_external.py`. The DermLIP best checkpoint is portable: `best.pt` (345 MB) reproduces every number in §2.2 from the saved internal-test split.